

RECONCILIATION AUDIT
REPORT

Evidence-backed verification of the active land dataset

STATUS: REVIEW REQUIRED

AREA: Mumbai — Andheri East

1. DATASET INFORMATION

Dataset Name:	Mumbai — Andheri East	Dataset ID:	mumbai-andheri
City / Urban Area:	Mumbai (Andheri East)	Dataset Version:	v1.2
Processing Date:	2026-09-03	Authoritative CRS:	EPSG:32643 (UTM 43N) → EPSG:4326
Total Entities:	72 Parcels	Verification State:	REVIEW REQUIRED

2. RECONCILIATION SUMMARY

Metric Category	Parcel Count	Distribution (%)	Audit Status
Verified (Auto-Harmonized)	52	72.2%	VERIFIED · READY
Surveyor Review Required	14	19.4%	REVIEW RECOMMENDED
Spatial Conflict Detected	6	8.3%	CONFLICT QUEUED
Total Land Parcels Analyzed	72	100.0%	Avg Confidence: 89.6%

3. SOURCE EVIDENCE

Authoritative Data Source	Layer Type	Participating Parcels	Coverage	Status
City Survey Cadastre (CTS)	Cadastral	72 parcels	100.0%	VERIFIED
BMC Property Tax Directorate	Municipal	72 parcels	100.0%	VERIFIED
High-Res Aerial Orthophoto	ORI	72 parcels	100.0%	VERIFIED
LOD2 Segmentation Model	AI	72 parcels	100.0%	VERIFIED
Survey of India CORS RTK Benchmarks	RTK	8 parcels	11.1%	VERIFIED

4. SPATIAL QUALITY BENCHMARKS

Quality Measure	Measured Value	Authoritative Threshold	Audit Status	Evaluation Purpose
IoU	90.7% (0.907)	≥ 85.0%	PASSED	Measures how closely source boundaries overlap with the reconciled boundary.
Boundary Agreement	89.6% (0.896)	≥ 80.0%	PASSED	Measures consistency between available source boundaries.

Ground Truth Agreement	89.3% (0.893)	≥ 80.0%	PASSED	Measures agreement with verified survey or ground-reference data.
Verification Confidence	89.6% (0.896)	≥ 85.0%	PASSED	Shows the current confidence assigned to the reconciliation result.

5. SPATIAL CONFLICT SUMMARY

CONFLICT CLASSIFICATION	INCIDENCE COUNT	DISCREPANCY DESCRIPTION	RESOLUTION STATUS
Boundary Offset	9	Lateral coordinate discrepancy between cadastral boundary and building edge	PENDING REVIEW
Source Mismatch	6	Area or geometry shape disagreement between revenue deeds and municipal GIS	PENDING REVIEW
Infrastructure Crossing	4	Proximity or spatial overlap with designated road ROW or SWD buffer	PENDING REVIEW
Geometry Issue	1	Self-intersection, sliver polygon, or vertex collinearity anomaly	PENDING REVIEW

6. PARCEL RECONCILIATION LEDGER (ISSUES REQUIRING INSPECTION)

The following parcels exhibit coordinate boundary drift, lower consensus overlap, or multi-source discrepancies requiring cadastral surveyor review:

PARCEL ID	SURVEY NO	CONFLICT TYPE	SOURCES	DRIFT	IOU	CONF	STATUS
MUM-202	CTS-1401/1	Source Mismatch	3 Sources	1.06m	0.87	77%	REVIEW
MUM-204	CTS-1403/A	Source Mismatch	3 Sources	3.4m	0.65	61%	CONFLICT
MUM-207	CTS-1406/A	Infrastructure Crossing	3 Sources	0.9m	0.83	82%	REVIEW
MUM-208	CTS-1407/1	Geometry Issue	3 Sources	3.2m	0.75	65%	CONFLICT
MUM-212	CTS-1411	Infrastructure Crossing	3 Sources	0.98m	0.88	87%	REVIEW
MUM-213	CTS-1412/A	Boundary Offset	3 Sources	2.9m	0.68	59%	CONFLICT
MUM-217	CTS-1416/1	Boundary Offset	3 Sources	0.9m	0.81	80%	REVIEW
MUM-222	CTS-1421/A	Boundary Offset	3 Sources	1.46m	0.82	85%	REVIEW
MUM-226	CTS-1425/1	Source Mismatch	3 Sources	3.4m	0.65	61%	CONFLICT
MUM-227	CTS-1426	Infrastructure Crossing	3 Sources	1.54m	0.81	78%	REVIEW
MUM-232	CTS-1431/1	Boundary Offset	3 Sources	1.14m	0.86	83%	REVIEW
MUM-237	CTS-1436/A	Infrastructure Crossing	3 Sources	1.22m	0.81	76%	REVIEW
MUM-242	CTS-1441	Boundary Offset	3 Sources	2.1m	0.72	66%	CONFLICT
MUM-247	CTS-1446/1	Boundary Offset	3 Sources	1.14m	0.88	86%	REVIEW
MUM-252	CTS-1451/A	Boundary Offset	3 Sources	1.62m	0.82	79%	REVIEW

7. EVIDENCE COVERAGE & AUDIT PILLARS

EVIDENCE DOMAIN	COVERAGE STATE	PROVENANCE & METHODOLOGY NOTE
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Geometry Validation	COMPLETE	All polygon boundaries checked for closure, orientation, and valid topology
CRS Validation	COMPLETE	Source geometries transformed to EPSG:32643 (UTM 43N) → EPSG:4326 with zero angular distortion
Source Evidence	COMPLETE	5 authoritative source layers participating with immutable provenance
Ground Truth	COMPLETE	8 CORS / RTK GNSS benchmark survey points correlated
Historical Evidence	NOT AVAILABLE	Prior year cadastral revision snapshots attached for temporal comparison
Infrastructure Evidence	COMPLETE	Road Right-of-Way and Stormwater Drainage corridor crossing evaluation active

8. RECONCILIATION PIPELINE PROGRESSION

STAGE	PIPELINE OPERATIONAL STAGE	PROCESSING STATE	STAGE DESCRIPTION
01	INGEST (01 Multi-Source Ingestion)	COMPLETED	Cadastral, Municipal & Drone ORI package validation
02	AI_02B (02B Mask R-CNN Inference)	COMPLETED	Drone ORI footprint instance segmentation (ResNet-50-FPN)
03	AI_02C (02C Polygon + CRS)	COMPLETED	Contour vectorization & metric affine georeferencing
04	STAGE_03 (03 Schema Normalization)	COMPLETED	Triple-source convergence (Cadastral, Municipal, AI Source)
05	MATCH (07-08 STRtree Matching)	COMPLETED	R-tree candidate filtering & multi-signal matching
06	COMPARE (09 IoU & Drift Evaluation)	COMPLETED	IoU, centroid drift, and area delta comparison
07	VERIFY (10-12 Consensus & Review)	REVIEW	Ground truth and policy threshold verification
08	REGISTER (13-16 Authoritative Register)	NOT STARTED	Final legal cadastre registry commit

9. AUDIT HISTORY & LIFECYCLE EVENTS

EVENT TIMESTAMP	ACTION / LIFECYCLE EVENT	ACTOR / SUBSYSTEM	STATUS
2026-09-03 10:14:22	Dataset package imported and validated	Data Ingestion Engine	COMPLETED
2026-09-03 10:15:08	Spatial CRS projected to metric UTM and normalized	CRS Safety Engine	COMPLETED
2026-09-03 10:16:45	8-signal multi-source candidate spatial indexing completed	STRtree Matcher	COMPLETED
2026-09-03 10:18:12	Autonomous multi-source boundary consensus evaluated	Consensus Engine	COMPLETED
2026-09-03 10:19:30	20 parcels queued for surveyor inspection	State Machine	REVIEW REQUIRED
2026-09-03 10:21:04	Survey of India CORS RTK ground-truth checkpoints attached	GNSS Verifier	COMPLETED

10. CURRENT DATASET STATUS & SPATIAL VERIFICATION STATEMENT

The active dataset **Mumbai — Andheri East** (Mumbai — Andheri East) currently holds an audit status of **REVIEW REQUIRED**. A total of 52 of 72 parcels (72.2%) meet the strict dual-threshold criteria ($\text{IoU} \geq 0.85$ and Centroid Drift < 2.0 meters) and are certified for authoritative land record workflows. The remaining 20 parcels have been formally queued in the Conflict Ledger for surveyor inspection.

STATUTORY NOTICE: This document provides spatial geometry reconciliation, multi-source sensor agreement, and technical evidentiary verification only. It does not constitute a final judicial determination of legal title or ownership. Administrative registration of final boundaries must follow statutory procedures stipulated under the relevant State Land Revenue Act.

11. DATA SOURCES & RECONCILIATION METHODOLOGY

Methodological Architecture: Spatial reconciliation in TERRANODE employs an 8-signal scoring matrix (combining Intersection-over-Union, centroid offset, area delta, perimeter agreement, shape similarity, and bounding-box overlap) indexed with Shapely STRtrees. Coordinate references are validated in native projected UTM metric systems prior to comparison, eliminating angular degree distortion. Ground-truth verification uses Survey of India CORS GNSS stations as authoritative reference coordinates.